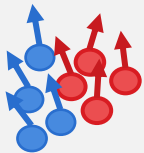


GRPO

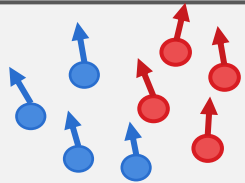
Semantic Diversity	✗
Gradient Diversity	✗
Direction-aware	✗



Reasoning  Memorization

EVOL-RL

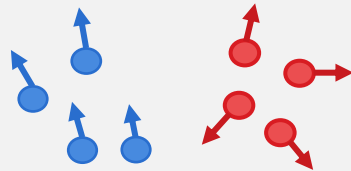
Semantic Diversity	✓
Gradient Diversity	✗
Direction-aware	✗



Reasoning  Memorization

G²RL

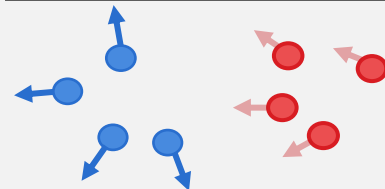
Semantic Diversity	✓
Gradient Diversity	✓
Direction-aware	✗



Reasoning  Memorization

DiRL

Semantic Diversity	✓
Gradient Diversity	✓
Direction-aware	✓



Reasoning  Memorization